

OBJECT DETECTION Using ESP32 Cam

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An ESP32 camera (cam) can take pictures at 2MB per second rate and transmit them up to 200 metres using a small an-

tenna on 2.4GHz. But when it teams up with an internet-enabled network it can do artificial intelligence (AI) related work as well.

In this project the ESP32 cam takes and sends the pictures to a remote computer that is connected to the internet, pulls the specified AI model, and analyses the object (model) to identify it. It then sends the object's identity back to the ESP32 cam, which then executes the object-specific command entrusted to it.

When the detection computer is on, the ESP32 cam is all vigilant and ready for action. Once the model is

loaded (identified), an active internet connection is no longer required. However, the computer needs to remain on all the time.

The ESP32 cam also reproduces the live feed from the ESP32 camera. The IP address is set as fixed type (192.168.43.11), which can be changed as per requirement.

The ESP32 continuously draws the image boundary over the image. If it finds two images, it draws two boundaries, and so on. Its confidence level can be set from 0 to 100%.

ImageResolution, MirrorImage, Flash (on 100% switch on the in-built LED), Quality, Brightness, and Contrast (see Fig. 6) can be changed. However, on increasing image resolution from 320×240 to 1600×1200 the system become extremely slow. It is better to leave the control at the default level and change only if necessary. Author's prototype is shown in Fig. 1.

Circuit and working

Object detection by ESP32 cam circuit is shown in Fig. 2. The project undertaken by the author as a test project was to monitor a road intersection or gathering of many people at a place in a restricted area.

The ESP32 cam detects the people gathering and sets the GPIO14 output as high. A switching transistor triggers input to a monostable delay timer (0 to



Fig. 1: Author's prototype

BILL OF MATERIAL	
Components	Quantity
ESP32 cam	1
IC555 (MOD3)	1
BC547 (T1, T2)	2
1k resistor (R4)	2
100k variable resistor (R2)	1
Capacitor 10nF (C2), 470μF (C3), 100μF (C1)	2
Voltage regulator LM1117 (MOD2)	1
Diode (D1)	
5V relay module (RL1)	1

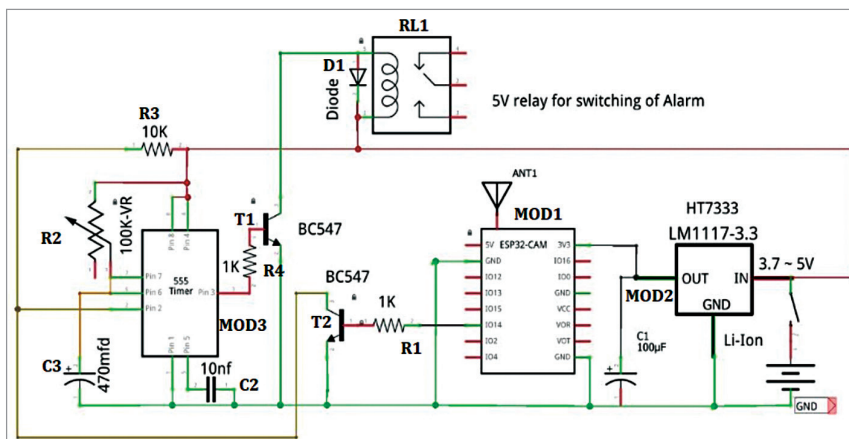


Fig. 2: Circuit diagram

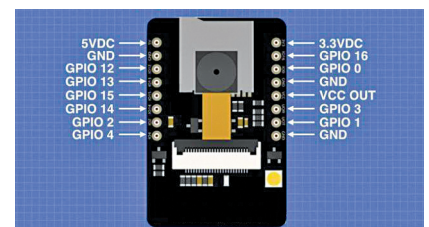


Fig. 3: Pin details of ESP32 cam